Solutions, Second Annual ECC Undergraduate Mathematics Competition, April 10, 1999

1. Multiples of 11.

By Fermat's little theorem, $2^{10} \equiv 1 \pmod{11}$, and hence $2^{1990} \equiv 1 \pmod{11}$, so $2^{1999} \equiv 2^9 \pmod{11}$. Since $2^5 = 32 \equiv -1 \pmod{11}$ and $2^4 \equiv 5 \pmod{11}$ we have $2^9 \equiv -5 \pmod{11}$ and thus $2^{1999} + 5 \equiv 0 \pmod{11}$.

SECOND SOLUTION

Since $2^5 = 32 \equiv -1 \pmod{11}$, for every odd integer k, we have $2^{5k} \equiv -1 \pmod{11}$, and in particular $2^{1995} \equiv -1 \pmod{11}$. As above, $2^4 \equiv 5 \pmod{11}$, so $2^{1999} + 5 \equiv 0 \pmod{11}$.

2. Pattern in squares.

One general formula is

$$n^2 + (n+1)^2 + n^2(n+1)^2 = (n^2 + n + 1)^2.$$

Here is a proof.

$$\begin{aligned} n^2 + (n+1)^2 + n^2(n+1)^2 &= n^2 + (n^2 + 2n + 1) + n^4 + 2n^3 + n^2 \\ &= n^4 + 2n^3 + 3n^2 + 2n + 1 \\ &= (n^2 + n + 1)^2; \end{aligned}$$

just multiply out and check.

3. Any rationals here?.

No, y_n is always irrational. For proof, note that given any n, y_n will contain arbitrarily long strings of 0s, introduced by the terms $(10^k)^n$, $k = 1, 2, 3, \ldots$. But the decimal expansion of a rational number is always ultimately repeating. If the sequence of digits which repeats has length m, then every string of 0s has length less than m, unless the decimal terminates (followed by an infinite string of zeros). But every contribution x^n in y_n begins with a nonzero digit, so no y_n is a terminating decimal.

4. Which is greater.

The second is the larger. We will show more generally that for $1/2 \le x < y < 1$ and $n \ge 2$, $x^n + (1-x)^n < y^n + (1-y)^n$. Since $1/2 < 2/3 < 1/\sqrt{2} < 1$, the claim follows.

Let $f(x) = x^n + (1-x)^n$, for $1/2 \le x \le 1$. Then $f'(x) = n[x^{n-1} - (1-x)^{n-1}] > 0$ when $x > 1/2$, and the assertion follows.

5. Approximating an integral.

The answers are: (a) $x_1 = -1/2$ and $x_2 = 1/24$; and (b) $71/192$.

(a) Write the polynomial as $f(x) = ax^2 + bx + c$. We seek x_1 and x_2 such that

$$\frac{a}{3} + \frac{b}{2} + c = ax_1^2 + bx_1 + c + 2ax_2 + b$$

holds, independent of a, b and c. Thus, we want

$$\frac{a}{3} = a(x_1^2 + 2x_2),$$

and

$$\frac{b}{2} = b(x_1 + 1).$$

The second equation is satisfied when $x_1 = -1/2$, and then the first holds as well when $x_2 = 1/24$.

(b) Let $g(x) = x^3 + ax^2 + bx + c$. Then $g(x) = x^3 + f(x)$, where $f(x)$ is as in (a). We have

$$\int_0^1 g(x)dx = \frac{1}{4} + \int_0^1 f(x)dx,$$

and

$$\begin{aligned} g(x_1) + g'(x_2) &= f(x_1) + f'(x_2) + x_1^3 + 3x_2^2 \\ &= \int_0^1 f(x)dx + \left(-\frac{1}{2}\right)^3 + 3\left(\frac{1}{24}\right)^2 \\ &= \int_0^1 g(x)dx - \frac{1}{4} - \frac{1}{8} + \frac{1}{192}, \end{aligned}$$

so the error is

$$\frac{1}{4} + \frac{1}{8} - \frac{1}{192} = \frac{71}{192} \doteq 0.37.$$

6. Estimating a sum.

$$\begin{aligned}\frac{k^2-2}{k!} &= \frac{k(k-1)}{k!} + \frac{k}{k!} - \frac{2}{k!} \\ &= \frac{1}{(k-2)!} + \frac{1}{(k-1)!} - \frac{2}{k!}.\end{aligned}$$

Thus,

$$\begin{aligned}S &= \sum_{k=2}^{100}\left(\frac{1}{(k-2)!} + \frac{1}{(k-1)!} - \frac{2}{k!}\right) \\ &= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{98!} \\ &\quad + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{98!} + \frac{1}{99!} \\ &\quad - \frac{2}{2!} - \cdots - \frac{2}{98!} - \frac{2}{99!} - \frac{2}{100!} \\ &= 1 + 2 - \frac{1}{99!} - \frac{2}{100!} < 3.\end{aligned}$$

SECOND SOLUTION

Let

$$d_n = 3 - \sum_{r=2}^{n} \frac{r^2-2}{r!}.$$

Then $d_2 = \frac{4}{2!}$, and examination of further d_n suggests that $d_n = \frac{n+2}{n!}$. We prove this by induction. We have it for $n = 2$. Suppose that $d_k = \frac{k+2}{k!}$. Then

$$\begin{aligned}d_{k+1} &= \frac{k+2}{k!} - \frac{(k+1)^2-2}{(k+1)!} \\ &= \frac{(k+1)(k+2) - (k+1)^2 + 2}{(k+1)!} \\ &= \frac{k^2+3k+2-k^2-2k-1+2}{(k+1)!} \\ &= \frac{(k+1)+2}{(k+1)!},\end{aligned}$$

completing the induction. Thus every $d_n > 0$. In particular, $d_{100} > 0$, and therefore $S < 3$. ∎

7. Close to an integer.

Let $s_n = a_1 + a_2 + \cdots + a_n = b_n + r_n$, where b_n is an integer and $0 \le r_n < 1$. If we partition the interval $[0,1)$ into 1000 subintervals $[0,.001)$, $[.001,.002), \ldots, [.999,1)$, each of the 1999 numbers r_n falls into one of these subintervals, and since there are more r_ns than subintervals, some subinterval contains at least two of them; say r_n and r_{n+k} lie in the same subinterval. Then

$$\begin{aligned} a_{n+1} + \cdots + a_{n+k} &= s_{n+k} - s_n \\ &= b_{n+k} - b_n + r_{n+k} - r_n, \end{aligned}$$

which differs from the integer $b_{n+k} - b_n$ by $|r_{n+k} - r_n|$, which is less than .001.

8. Lots of nines.

We have $a_0 = 10 - 1$, and by direct calculation we find $a_1 = 22599 = 226 \cdot 10^2 - 1$. We prove by induction that a_n always has the form $b_n \cdot 10^{2^n} - 1$, where b_n is a positive integer. It will follow that a_n in decimal form ends in 2^n nines, and in particular that a_{10} ends in $2^{10} = 1024$ nines.

The cases $n = 0$ and $n = 1$ are already verified. Suppose that

$$a_k = b_k \cdot 10^{2^k} - 1$$

for some positive integer b_k. Then

$$\begin{aligned} a_{k+1} &= 3(b_k \cdot 10^{2^k} - 1)^4 + 4(b_k \cdot 10^{2^k} - 1)^3 \\ &= 3(b_k^4 \cdot 10^{4\cdot 2^k} - 4b_k^3 \cdot 10^{3\cdot 2^k} + 6b_k^2 \cdot 10^{2\cdot 2^k} - 4b_k \cdot 10^{2^k} + 1) \\ &\qquad + 4(b_k^3 \cdot 10^{3\cdot 2^k} - 3b_k^2 \cdot 10^{2\cdot 2^k} + 3b_k \cdot 10^{2^k} - 1) \\ &= 3b_k^4 \cdot 10^{4\cdot 2^k} - 8b_k^3 \cdot 10^{3\cdot 2^k} + 6b_k^2 \cdot 10^{2\cdot 2^k} \qquad\qquad - 1 \\ &= b_{k+1} \cdot 10^{2^{k+1}} - 1, \end{aligned}$$

where $b_{k+1} = 3b_k^4 \cdot 10^{2\cdot 2^k} - 8b_k^3 \cdot 10^{2^k} + 6b_k^2$.

It is clear that b_{k+1} is an integer, and since all a_k are clearly positive, it is clear also that b_{k+1} is positive. The desired result now follows by induction.

9. Roots of a cubic.

(a) Let $f(x) = ax^3 + bx + c$. That r is a double root implies that both $f(r)$ and $f'(r)$ are zero. Thus

$$f(r) = ar^3 + br + c = 0, \quad \text{and}$$
$$f'(r) = 3ar^2 + b = 0.$$

If we multiply the first equation by 3, the second equation by r, and subtract we obtain

$$2br + 3c = 0,$$

from which we have immediately that $r = -3c/2b$, q.e.d.

(b) From the solution to (a) we see that since b and c are rational, so is r rational. Let s be the third root, and recall that $a \neq 0$. Then

$$\begin{aligned} x^3 + \frac{b}{a}x + \frac{c}{a} &= (x-r)^2(x-s) \\ &= x^3 - (2r+s)x^2 + (r^2+2rs)x - r^2s. \end{aligned}$$

Upon equating coefficients we have $s = -2r$ and

$$\begin{aligned} \frac{b}{a} &= r^2 + 2rs \\ &= r^2 - 4r^2 \\ &= -3r^2, \end{aligned}$$

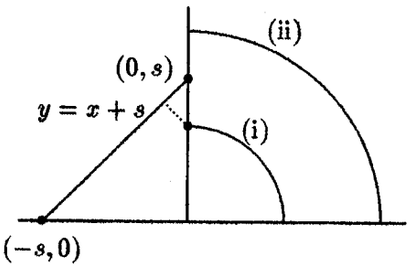
so that

$$|r| = \sqrt{\frac{-b}{3a}}.$$

We have already observed that r is rational, so the result follows.

10. Minimization.

$$\text{The minimum value is } \begin{cases} \dfrac{(r-s)^2}{2} & \text{if } r < s \\ (r-s)^2 & \text{if } r \geq s. \end{cases}$$



The quantity $E := (a+d)^2+(b-c)^2$ is the square of the distance from $P = (a,b)$ to $Q = (-d,c)$. The point P lies in the first quadrant portion of the circle $x^2 + y^2 = r^2$ and Q lies on the second quadrant portion of the line $y = x+s$. There are two cases (see figure at left): (i) If $r < s$, as in the smaller circle, E is minimized by choosing $P = (0,r)$ and $Q = ((r-s)/2, (r+s)/2)$, so that $\overline{PQ}$ has slope -1. The minimum value of E then is $E = (s-r)^2/2$. (ii) If $r \geq s$ then E is minimum when $P = (0,r)$ and $Q = (0,s)$, where $E = (r-s)^2$. ∎

ALGEBRAIC SOLUTION

Again let

$$E = (a+d)^2 + (b-c)^2 = r^2 + d^2 + c^2 + 2ad - 2bc.$$

Now, $r \geq b$, so

$$rc \geq bc \geq bc - ad; \quad \text{i.e. } 2ad - 2bc \geq -2rc \tag{1}$$

Therefore

$$\begin{aligned} E &\geq r^2 - 2rc + c^2 + d^2 \\ &= (r-c)^2 + d^2 \\ &= (r-c)^2 + (s-c)^2. \end{aligned} \tag{2}$$

Let $F(c) = (r-c)^2 + (s-c)^2$ for $0 \leq c \leq s$. This quadratic has its minimum value at $c = (r+s)/2$, which is in $[0,s]$ if $r \leq s$. Thus: (i) If $r \leq s$, then $F(c) \geq F((r+s)/2) = (r-s)^2/2$. With $a = 0$ and $b = r$ we have equality in (1) and (2), so E has minimum value $(r-s)^2/2$, occurring when $c = (r+s)/2$, $d = (s-r)/2$, $a = 0$ and $b = r$. (ii) If $r > s$, then $f'(c) = 2(2c - r - s) < 2(2c - 2s) < 0$, so $F(c)$ decreases for $0 < c < s$ and has minimum value when $c = s$, $d = 0$. Again we have equality in (2) when $a = 0$ and $b = r$, so E has minimum value $(r-s)^2$. ∎